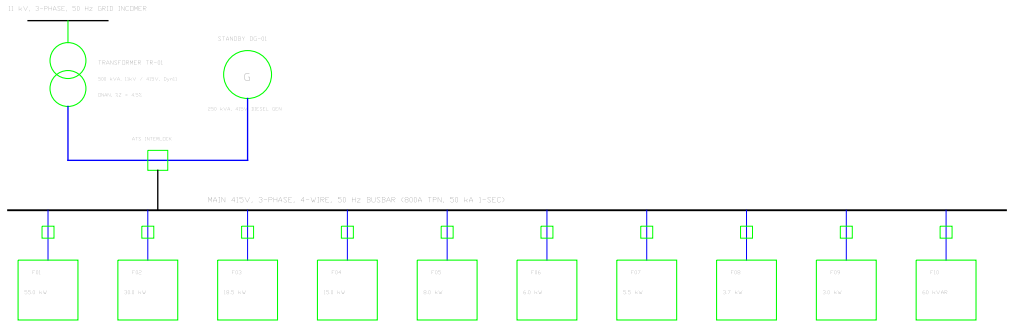


ELECTRICAL LOAD SUMMARY, TRANSFORMER SIZING & POWER FACTOR CORRECTION

1. Total Plant Connected Load: 1843 kW | Operating Diversity Factor: 0.70 | Maximum Running Demand: 1292 kW.
2. Operating Power Factor: 0.83 (uncorrected) → Corrected to 0.98 lag via 60 kVAR Automatic Power Factor Capacitor Bank.
3. Apparent Power Demand: 1318 kVA @ 0.98 PF | 500 kVA Transformer Utilization: 26.4% (Generous margin for phase 2 expansion).
4. Standby Generator (DG-01): 250 kVA (200 kW @ 0.8 PF) rating provides 100% emergency running capacity during grid outage.
5. Motor Protection: Class 10/20 bimetallic overload relays, phase failure protection, thermistor inputs for blower motors.
6. Earthing System: Combined TN-S earthing grid with dedicated chemical earthing pits (<1.0 ohm overall resistance).
7. Lightning Protection: Early Streamer Emission (ESE) air terminal per IS/IEC 62305 with 45m protection radius.

CONCEPTUAL FEED – PRELIMINARY – NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT

HARAKUTIR AGRO-PROCESS ENGINEERING PVT. LTD. 15,000 kg/day Solid Pigeon Pea Optimization Greenfield Project		
ELECTRICAL SINGLE-LINE DIAGRAM & 415V MCC POSITIVE ELECTRICAL 3-STAGE CONCEPTUAL FEED		
DWG NO: DWG-FIN-1ST-011	REV: V4.0	SHEET: 11 OF 12
SCALE: AS SHOWN DATE: 2026-11-10	STATUS: PRELIMINARY	DATE: 0-4 YEILD